

No. of Functional Features Included in the Solution

Date	30 June 2026
Team ID	XXXXXX
Project Name	OptiCrop – Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status	Marks Contrib .
1	Crop Recommendation Engine	Accepts soil & climate inputs (N,P,K, temp, humidity, pH, rainfall) and recommends the best crop using ML.	Core ML Module	Done	1.0
2	Soil-Climate Suitability Checker	Allows user to select a crop and checks if current soil/climate parameters are suitable for that crop.	ML Module	Done	0.8
3	Fertilizer Optimization Module	Recommends optimal N, P, K fertilizer quantities based on crop type to minimize waste and cost.	Backend Service	Done	0.8
4	Farmer Input Dashboard	Web-based UI form for farmers to enter soil parameters and view crop recommendations easily.	Presentation Layer	Done	0.7
5	Researcher Analytics Portal	Dashboard showing aggregate crop-environment charts, regional patterns, and production trend insights.	Analytics Module	In Progress	0.5
6	Crop Compatibility Report	Generates a detailed report showing which parameters match/deviate from ideal ranges for a chosen crop.	Backend Service	Done	0.5
7	Historical Yield Tracker	Stores and displays past recommendations and farmer-entered yield outcomes for seasonal comparison.	Database Module	In Progress	0.4
8	REST API Endpoints	Flask-based REST API exposing crop recommendation, suitability, and fertilizer endpoints for integration.	API Gateway	Done	0.3

Feature Summary:

Metric	Count / Value
Total Features Planned	8

Total Features Implemented	6 (Done) + 2 (In Progress)
Core / Must-Have Features	5 (Crop Recommendation, Suitability Checker, Fertilizer Module, Dashboard, API)
Additional / Nice-to-Have Features	3 (Analytics Portal, Compatibility Report, Yield Tracker)
Features Tested & Verified	6

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	2	Farmer Input Dashboard, Crop Compatibility Report UI
2	Backend / Logic	3	Crop Recommendation Engine, Suitability Checker, Fertilizer Optimizer
3	Database / Storage	1	Historical Yield Tracker (SQLite / PostgreSQL)
4	API / Integration	1	Flask REST API Endpoints
5	Analytics / Reporting	1	Researcher Analytics Portal